Ask or Commit? Clarification in Multimodal Reference Games

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Abstract

This paper investigates clarification efficiency in multimodal reference games , addressing the tendency of modern LLMs and VLMs to over-ask or over-commit. We propose a zero-shot expected-utility framework where a listener triggers clarification unless a confidence threshold $\max_i P(t_i|u) \geq 1 - c$ is met. Evaluations on synthetic scenes demonstrate that elaborate reasoning techniques, like Chain-of-Thought, paradoxically degrade cost-penalized accuracy by generating diffuse posteriors that trigger unnecessary questions. Ablation studies reveal that removing clarification entirely improves performance for most models. Furthermore, structured speaker constraints aid weaker models but interact non-monotonically with stronger ones, with most of the strong-speaker degradation arising from listener-side compatibility mismatch rather than intrinsic loss of speaker, side redundancy. Error analysis confirms the predominant failure mode is miscalibrated uncertainty, requesting clarification when the correct target is already ranked first—rather than poor grounding. Ultimately, uncertainty calibration under ambiguity remains the critical bottleneck for pragmatic agents.

1 Introduction

Reference games provide a fundamental testbed for pragmatic language understanding: a speaker must describe a target object among distractors, and a listener must infer the intended referent from the utterance. In ambiguous settings, listeners face a central decision problem: whether to commit to an interpretation or ask a clarifying question. As Large Language Models (LLMs) and Vision-Language Models (VLMs) are increasingly deployed in interactive systems, this trade-off becomes practically important. Existing systems often either over-commit to uncertain predictions or over-ask for clarification, reducing interaction efficiency.

In this work, we study clarification behavior in multimodal reference games through a simple decision-theoretic framework. Given a posterior distribution $P(t \mid u)$ over candidate objects conditioned on an utterance u, the listener compares the expected utility of committing against the utility of asking a clarification question with cost c. The resulting ask-or-commit rule reduces to a confidence threshold:

$$\max_i P(t_i \mid u) \geq 1 - c.$$

This framework provides a training-free mechanism for controlling clarification behavior while exposing how different prompting and inference strategies shape posterior uncertainty.

Using controlled synthetic scenes with varying levels of distractor overlap, we evaluate a range of LLM/VLM speaker and listener strategies in a zero-shot setting. Surprisingly, we find that more elaborate reasoning strategies consistently worsen interaction efficiency. In particular, Chain-of-Thought (CoT) prompting substantially increases clarification rates by producing diffuse or bimodal posterior distributions, even when the underlying top prediction is correct. In contrast, simpler listeners that make direct hard decisions often achieve substantially higher cost-penalized accuracy.

Our results suggest that the primary bottleneck in modern pragmatic agents is not insufficient reasoning capability, but miscalibrated uncertainty under ambiguity. More broadly, the proposed framework provides a controlled setting for studying the relationship between pragmatic inference, confidence calibration, and interactive decision making in LLM-based systems.

2 Literature Review / Related Work

Pragmatic reasoning and reference games. Pragmatic reasoning has long been studied in computational linguistics and cognitive science through the Rational Speech Act (RSA) framework, which models communication as recursive probabilistic inference between speakers and listeners (Fried et al., 2018; Cohn-Gordon et al., 2018). Reference games and referring expression generation provide natural benchmarks for evaluating such behavior, and recent work extends pragmatic inference to multimodal settings including contrastive captioning and structured signaling environments (Ou et al., 2023; Carlsson and Dubhashi, 2023). However, classical RSA typically assumes a discrete utterance space with hand-specified semantics, making it difficult to apply directly to the open-ended outputs of modern LLMs and VLMs.

LLMs, uncertainty, and clarification. Recent work investigates whether LLMs behave as pragmatic communicative agents (Jian and Siddharth, 2024; Gul and Artzi, 2024). Prior approaches often learn clarification strategies through supervised interaction or self-play, explicitly optimizing the trade-off between task success and clarification cost (Berant et al., 2025). Other work studies pragmatic failures in multimodal generation, including over-verbose referring expressions and hallucinated attributes in VLM systems (Ma et al., 2025). In contrast to learned clarification policies, we study whether a simple expected-utility rule can effectively govern clarification behavior in a zero-shot setting. Our experiments further show that the main failure mode is often not incorrect reasoning itself, but poorly calibrated uncertainty estimates that trigger unnecessary clarification.

3 Proposed Method

3.1 Core Idea

The central question of this work is: *when should a listener ask a clarifying question rather than commit to an answer?*

Concretely, given a posterior $P(t \mid u)$ over N candidate objects, the listener's two actions have expected utilities:

$$\mathrm{EU}(\text{commit}) = \max_i P(t_i \mid u), \tag{1}$$

$$\mathrm{EU}(\text{ask}) = 1 - c, \tag{2}$$

where $c \in (0, 1)$ is the per-question cost. The listener commits when $\mathrm{EU}(\text{commit}) \geq \mathrm{EU}(\text{ask})$, i.e.

$$\max_i P(t_i \mid u) \;\geq\; 1 - c. \tag{3}$$

Figure 1 illustrates the full two-agent pipeline. The **speaker** is given the target object's attributes and produces a referring utterance u. The **listener** observes the scene and u, computes a posterior, and applies the EU rule. If the rule triggers asking, the listener emits a yes/no clarification question; an oracle answers; the listener updates its posterior and may ask again (up to two turns); it then commits.

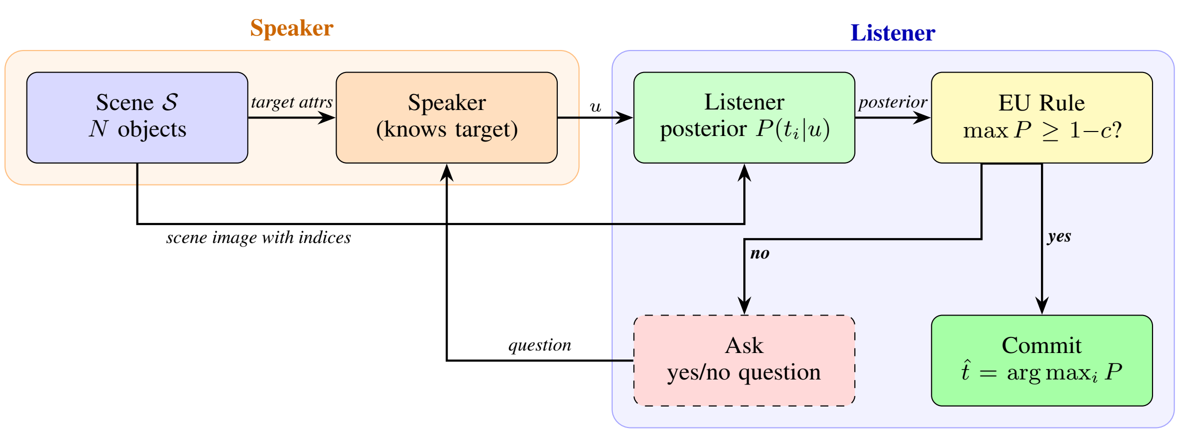

Figure 1: The cost-aware reference game pipeline. The speaker knows the target and generates a referring utterance u. The listener computes a posterior and applies the EU rule (Eq. 3).

3.2 TASK AND DATA

A **scene** $\mathcal{S} = \{o_1, \ldots, o_N\}$ contains N objects, each described by four discrete attributes:

1. `color` $\in$ {red, blue, green, yellow, purple, orange},
2. `shape` $\in$ {circle, square, triangle},
3. `size` $\in$ {small, medium, large}, and
4. a continuous 2-D position $(x, y) \in [0, 100]^2$ (bottom-left origin; y increases upward).

One object is designated the **target** $t \in \mathcal{S}$; all others are **distractors**. We generate scenes with $N \in \{6, 8, 10\}$ under two overlap conditions. In the `none` condition, there would be no spatial overlapping between the objects. In the `force` condition, objects are forced to be overlapping. Scene images are rendered at 512×512 pixels using PIL. (see Appendix 4) Each $(N, \text{condition})$ pair contains 100 fixed scenes; all speaker and listener strategies see the same 100 scenes, giving 600 unique scenes in total.

3.3 SPEAKER STRATEGIES

The speaker receives the target's attributes as plain text (not a visual annotation) to avoid color hallucination on the rendered image. We implement eight distinct strategies:

Speaker	**Description**
`feature_canonical`	Rule-based; enumerates color and shape, then adds size and/or location only if needed for uniqueness. No LLM call.
`natural`	LLM generates a free-form referring expression (e.g., "the green circle at the bottom left").
`pragmatic`	LLM is prompted to be maximally informative while remaining concise; produces RSA-style level-1 pragmatic descriptions.
`scene_first`	LLM receives the full distractor list and produces the shortest uniquely identifying description.
`listener_aware`	LLM is told the listener can see an annotated scene image; crafts descriptions that exploit spatial landmarks.
`landmark`	LLM identifies a salient landmark object and describes the target as a spatial relation to it (e.g., "to the left of the large red square").
`contrastive`	LLM identifies the most confusable distractor (foil) and produces a contrastive description highlighting the distinguishing feature.
`superlative`	LLM uses uniqueness or superlative descriptions ("the only triangle", "the largest red object").

Table 1: Speaker strategies.

All LLM-based speakers use `gemini-2.0-flash-001` via the OpenRouter API. Speaker utterances are cached per (scene, speaker) to avoid redundant API calls; once generated, the same utterance is reused across all listener evaluations.

3.4 LISTENER STRATEGIES AND POSTERIOR COMPUTATION

The listener receives the utterance u together with an index-annotated scene image. Its goal is to produce a probability distribution $P(t_i \mid u)$ over all N objects.

We implement six listener backends that differ in *how* they compute the posterior (Table 2). Five listeners (`index`, `direct`, `cot`, `elimination`, `feature-match`) directly emit a probability vector over object indices. The `coordinate` listener instead predicts a 2-D point $(\hat{x}, \hat{y})$ in scene coordinates, then converts the prediction to a posterior via a Gaussian distance kernel:

$$P(t_i \mid u) \;\propto\; \exp\!\left(-\frac{\|\,(\hat{x}, \hat{y}) - (x_i, y_i)\,\|^2}{2\sigma^2}\right), \quad \sigma = 10. \tag{4}$$

The bandwidth σ=10 (on the $[0, 100]^2$ grid) was chosen so that a prediction landing squarely on an object assigns ≥ 0.97 probability to that object when N=6; predictions that fall between two nearby objects produce a bimodal posterior that correctly reflects ambiguity. No temperature parameter is tuned; the kernel is fixed across all conditions.

Listener	**Description**
`index`	Single-pass: outputs a single index as the final answer.
`direct`	Outputs probabilities over indices.
`cot`	Chain-of-thought: first describes each indexed object, then scores each, then normalises.
`elimination`	Sequentially eliminates objects inconsistent with the utterance; posterior is uniform over survivors.
`feature_match`	Parses utterance into attribute tokens; soft-matches each token against object attributes to form a score vector.
`coordinate`	Predicts a 2-D coordinate $(\hat{x}, \hat{y})$ and converts to a posterior via Eq. 4.

Table 2: Listener strategies evaluated in this work.

When the EU rule triggers asking, the listener generates a yes/no clarification question which is posed directly to the speaker. The speaker answers deterministically from the ground-truth target attributes. The posterior is then updated by zeroing out objects inconsistent with the answer and renormalizing, and the EU rule is re-evaluated; this repeats for up to two turns before the listener commits.

3.5 Evaluation Metrics

Accuracy is the fraction of *all* n scenes in which the listener identifies the correct target. **Clarification Rate** (ask%) is the fraction of scenes where at least one question is asked. **Cost-Penalised Accuracy** (CPA) rewards correct answers and penalises questions:

$$\text{CPA} = \text{Accuracy} - c \times \text{ask\%}. \tag{5}$$

4 Experiment Results

We evaluate 28,800 trials over speaker×listener×scene-size×condition configurations at c=0.25.

4.1 Free-form speakers are more robust to scale than structured speakers, while landmark descriptions collapse

Figure 2 shows listener accuracy for all eight speakers with the `index` listener across $N \in \{6, 8, 10\}$ objects under both overlap conditions.

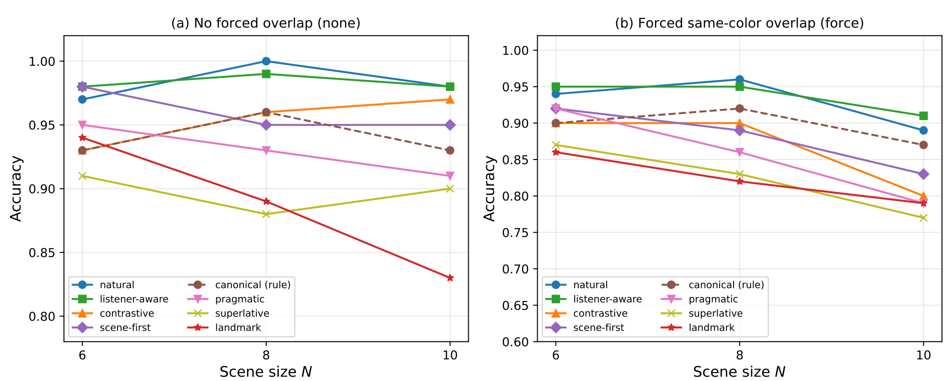

Figure 2: Accuracy vs. scene size for all eight speaker strategies, using the `index` listener at c=0.25. Left: no forced overlap (`none`); right: forced same-color distractor (`force`). Marker size is uniform; line styles distinguish speakers.

Speaker	N=6	N=8	N=10
`natural`	0.970	**1.000**	**0.980**
`listener_aware`	**0.980**	0.990	**0.980**
`contrastive`	0.930	0.960	0.970
`scene_first`	**0.980**	0.950	0.950
`feature_canonical`	0.930	0.960	0.930
`pragmatic`	0.950	0.930	0.910
`superlative`	0.910	0.880	0.900
`landmark`	0.940	0.890	0.830
Uniform baseline	0.167	0.125	0.100

Table 3: Speaker comparison with the `index` listener, c=0.25, `none` condition. Since ask% = 0 for all, CPA = Accuracy. Best per column in **bold**.

`natural` and `listener_aware` are the most robust speakers. Both maintain ≥ 0.97 accuracy at N=10 (none), with `natural` achieving a perfect 1.000 at N=8. These speakers generate concise, spatially grounded descriptions that uniquely identify the target even in dense scenes. The rule-based `feature_canonical` is competitive (0.93–0.96) with zero LLM calls, matching or exceeding LLM-based pragmatic and landmark speakers.

Landmark descriptions collapse at large N. `landmark` drops from 0.940 at N=6 to 0.830 at N=10 on the `none` condition ($\Delta = -0.110$), and further to 0.650 on the `force` condition at N=10. This failure is systematic: with more objects, a landmark that is itself described by color+shape becomes ambiguous (two yellow circles), creating a cascading reference failure. `superlative` shows a smaller but consistent degradation from $0.910 \to 0.900$.

Forced overlap is a clean difficulty dial. Under `force`, all speakers degrade (average $\Delta = -0.045$ at N=8), but the rank ordering is preserved: the best-performing speakers on `none` remain best on `force`.

4.2 Listener strategy determines the accuracy–cost trade-off; ask rate is the dominant CPA penalty

Figure 3 plots accuracy against ask rate for all listener strategies paired with the `feature_canonical` speaker. Each point is one (listener, N) configuration; iso-CPA lines are overlaid. *Ideal listeners occupy the top-left corner* (high accuracy, low ask).

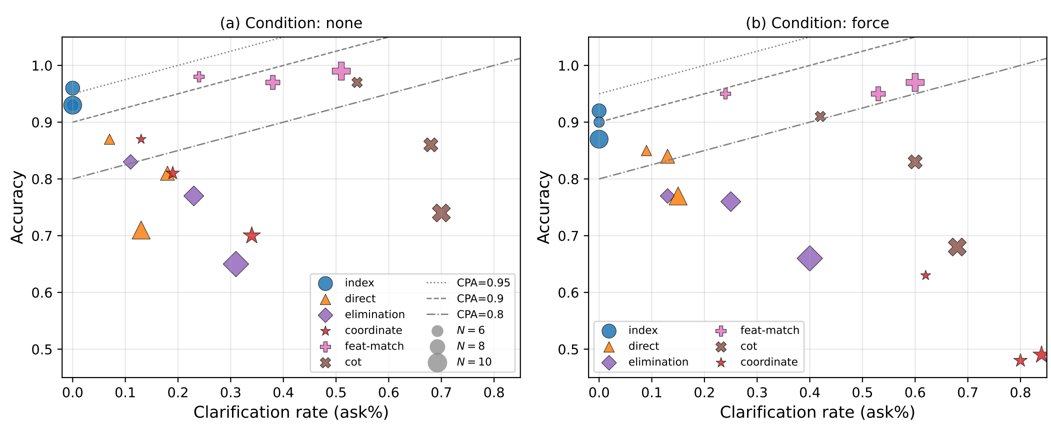

Figure 3: Accuracy vs. clarification rate for all listener strategies, `feature-canonical` speaker, c=0.25. Point size encodes scene size $N \in \{6, 8, 10\}$. Dashed/dotted lines are iso-CPA contours.

The `index` listener dominates in CPA by never asking. Its structured attribute-table input eliminates all referential ambiguity in the text channel: the posterior saturates to $\max_i P \approx 1.0$ at

Listener	none			force		
	Acc	Ask%	CPA	Acc	Ask%	CPA
`index`	**0.960**	0.00	**0.960**	**0.920**	0.00	**0.920**
`direct`	0.810	0.18	0.765	0.840	0.13	0.807
`elimination`	0.770	0.23	0.713	0.760	0.25	0.698
`feature-match`	0.970	0.38	0.875	0.950	0.53	0.817
`coordinate`	0.810	0.19	0.763	0.480	0.80	0.280
`cot`	0.860	0.68	0.690	0.830	0.60	0.680

Table 4: Listener comparison with `feature-canonical` speaker, c=0.25, N=8.

every scene, so the EU rule never fires. CPA equals accuracy (0.960 at N=8, none), the highest of any listener.

CoT reasoning inflates ask rate to $> 60\%$, substantially reducing CPA. Despite achieving accuracy comparable to the top listeners (0.860 at N=8), the `cot` listener's CPA is only 0.690 because it asks on 68% of scenes. Each time the model verbalises visual uncertainty ("Object 3 appears either blue or green"), the resulting posterior approaches uniform, triggering the EU rule even when the underlying argmax would have been correct. This is the largest CPA gap of any listener strategy—a -0.270 drop from `index` to `cot`—and illustrates that expressed calibration harms the clarification rule.

`coordinate` degrades catastrophically under forced overlap. On `none`, `coordinate` achieves 0.810 accuracy and 0.763 CPA, comparable to `direct`. On `force`, accuracy falls to 0.480 and CPA drops to 0.280 as the ask rate rises to 80%. When a same-color distractor sits nearby, the predicted coordinate falls between target and foil, producing a flat Gaussian posterior that perpetually triggers clarification—but yes/no answers cannot resolve a coordinate prediction error. The `feature-match` listener shows a similar pattern: very high accuracy (0.950–0.970) but excessive asking (38%–53%), yielding CPA of 0.875/0.817 (none/force) well below its accuracy, confirming that posterior flatness need not correspond to genuine referential ambiguity.

5 Ablations

We analyze three components of the framework: the expected-utility (EU) clarification rule (and whether posterior calibration can rescue it), the relationship between listener uncertainty and clarification benefit, and the interaction between speaker model and prompt structure. All numbers are at c=0.25; full results in Table 5.

The EU rule hurts; calibration explains most but not all. Removing the rule (always commit to $\arg\max_i P(t_i \mid u)$, so CPA reduces to raw accuracy) improves CPA for every listener with a non-degenerate posterior; the chain-of-thought (CoT) listener gains the most (Δ=+0.589). The same pattern replicates with comparable magnitudes (Appendix A.3), so the failure is not a text-affordance artifact. Sharpening posteriors via temperature scaling $P_\tau(i) \propto P(i)^{1/\tau}$ at τ=0.25 closes the CoT EU-rule penalty from -0.589 to -0.143 — a 76% recovery — but no $\tau \in \{0.25, 0.5, 1.0\}$ flips the sign for any listener (Appendix A.2). Calibration accounts for most of the failure; a residual scene-selection misranking persists (next paragraph).

Listener entropy is anti-informative about clarification benefit. The residual is a misranking problem: the cached dialogue listener (depth r=2, allowed a clarifying question before committing) underperforms the no-asking index baseline by 7.6 CPA points pooled over 600 scenes, and on the 93 scenes it flagged as uncertain, accuracy collapses from 71.0% (no clarification) to 37.6% (after clarification) — a 33.4 pp drop. Listener entropy fails to identify scenes where clarification helps; in our cached records the highest-entropy scenes are precisely where clarification hurts most.

Speaker prompt structure interacts non-monotonically with model capacity. Crossing the speaker model (gemini-flash vs. claude-sonnet-4) with prompt format (*natural* vs. *feature-canonical*) on `scenes_8_force` (n=100): the canonical prompt helps the weak speaker (Δ_{flash}=+0.035) and hurts the strong one (Δ_{sonnet}=-0.175, dropping CPA from 0.865 to 0.690).

Component ablated	Slice	Full	Ablated	Δ
EU rule on $\rightarrow$ always-commit (*cot*)†	pooled, n=600	0.242	0.832	+0.589†
Posterior temperature τ=1.0 $\rightarrow$ τ=0.25 (*cot*, rule on)	pooled, n=600	0.242	0.689	+0.446
Clarification on (*dialogue*) $\rightarrow$ off (*index*)†	pooled, n=600	0.842	0.918	+0.076†
Speaker model gemini-flash $\rightarrow$ sonnet-4 (*canonical*)	`scenes_8_force`, n=100	0.950	0.690	−0.260

Table 5: Ablation summary at c=0.25. The EU-rule, temperature, and dialogue rows pool the six datasets with the *feature-canonical* speaker; the speaker-model row uses a *direct-rank*(gemini-flash) listener. The vocabulary-constraint result from the previous draft appears in Appendix A.4.
†Positive Δ means *removing* the asking-capable component (the EU rule, or the dialogue listener's clarification turn) improves CPA.

6 Error Analysis and Qualitative Demonstrations

Across the experiment, most errors come from miscalibrated uncertainty rather than failure to find the target: some listeners already rank the correct object first but still ask, and the cached dialogue listener — even when allowed to clarify before committing — fails to recover from this miscalibration. It underperforms the no-clarification index baseline by 7.6 CPA points and *worsens* accuracy from 71% to 38% on the very scenes it flags as ambiguous. These results collectively suggest that calibration quality, rather than reasoning sophistication, is the primary determinant of pragmatic behavior in modern LLM/VLM agents — and that current LLM listeners cannot productively use their own clarifications to escape this bottleneck.

Pattern	Quantitative signal	Representative example	Insight
Asks when the answer is already clear	CoT is right 76.5% of the time, yet still asks in 62.9% of cases.	*"The green circle at the bottom left"* — finds it, yet asks.	Sharper CoT calibration would turn correct guesses into commits.
Perceptual flatness in the listener	The listener is right 70.8% of the time, yet still asks in 48.0% of cases.	*"the medium green triangle"* — already top-1, yet it asks.	A sharper posterior would cut asks and raise CPA.
Description missing a key detail	Two objects fit the description, so probability splits 0.5/0.5.	*"the large green triangle"* — two objects fit, so it cannot choose.	Here asking helps: the description is genuinely underspecified.

Table 6: Combined error analysis. Most mistakes do not come from completely missing the target. More often, the model is either too unsure and asks unnecessarily, or it focuses on the wrong detail.

7 Limitations and Future Work

Our benchmark is intentionally controlled: scenes are synthetic, the attribute vocabulary is closed, and most targets can be described using only color, shape, size, and coarse location. This makes ambiguity measurable and reproducible, but it also removes many difficulties of real multimodal reference, such as cluttered backgrounds, subtle visual distinctions, occlusion, open vocabulary descriptions, and longer discourse context. A second simplification is the decision rule itself: the listener assumes a fixed clarification cost and treats a follow up question as if it would resolve uncertainty perfectly in expectation. These assumptions are analytically useful, but they compress the richer trade-offs of natural multi-turn interaction.

A natural next step is to separate *referential* uncertainty from *perceptual* uncertainty before applying the ask-or-commit rule. That could involve better posterior calibration, higher resolution or region cropped visual inputs, lightweight verifiers that check whether a candidate truly matches the utterance, and learned clarification policies that choose both whether to ask and what to ask. Beyond the synthetic setting, the framework should be evaluated on natural images, open-ended referring expressions, and human in the loop studies, where the value of clarification can be measured in usability terms rather than CPA alone.

8 Conclusion

Across 28,800 trials, the central lesson is that effective pragmatic behavior depends more on calibrated confidence than on elaborate reasoning prompts. Constrained speakers and hard decision listeners often outperform more verbose systems because they avoid confidence smearing and unnecessary clarification, while in our experiments even targeted clarification questions failed to deliver this benefit at scale, suggesting current LLM listeners cannot use their own clarifications productively. In this sense, the main bottleneck for modern LLM/VLM agents in reference games is not the absence of reasoning steps, but the mismatch between posterior confidence and the real source of ambiguity.

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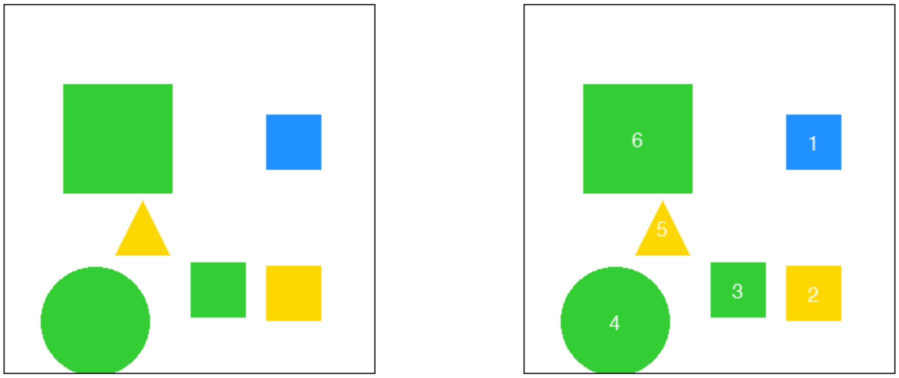
APPENDICES

REFERENCE GAME SCENE EXAMPLE



(a) Speaker's view — no labels (b) Listener's view — objects numbered

Figure 4: An example scene from `scenes_6_none`. The speaker sees the unlabeled scene and knows the target (object #4, large green circle, bottom-left); it must produce a referring expression. The listener sees the numbered scene and must select the correct index.

Speaker prompt (`listener_aware`):

```
You are a speaker in a visual reference game.

The TARGET object is: {target_description}

Your listener will see the SAME image, read your
description, and assign a
probability to each object being the one you mean. Your
goal is to write
a description that gives the listener maximum confidence in
the correct object.

Think: what description would make every OTHER object
clearly wrong, and this object clearly right? Prefer
descriptions using visually obvious properties
(color, shape, relative position) over subtle ones.

Output ONLY the referring expression (short, under 12
words). No other text.
```

Listener prompt (`CoTRankListener`, system message):

```
You are a listener in a visual reference game.

The image shows {n} objects labelled 1 to {n}.
Look at each numbered object carefully --- note its color,
shape, and size.

The speaker said: ``{utterance}''

Clarification cost: c = {cost_c}. You should commit with
confidence
or spread uncertainty honestly across candidates.

Reason step by step using only what you see:
1. OBSERVE --- for each label 1..{n}, describe what you see
(color, shape, size).
2. MATCH --- score how well the utterance describes each
object (0 = no match, 10 = perfect).
3. ASSIGN --- convert scores to probabilities summing to
1.0.

Output format (follow exactly):
```

```
SCORES: [s0, s1, ...]    ← integers 0--10
PROBS:  [p0, p1, ...]    ← floats summing to 1.0
```

A.1 EU RULE ROBUSTNESS ACROSS COST REGIMES

The cost-aware clarification rule's harm is preserved across clarification costs $c \in \{0.25, 0.5\}$ for every listener with a non-degenerate posterior. Pooled over the six datasets (n=600 per listener, *feature-canonical* speaker):

	c=0.25				c=0.5			
Listener	on	off	Δ	ask%	on	off	Δ	ask%
direct	0.739	0.808	-0.070	12.5%	0.772	0.808	-0.036	4.8%
cot	0.242	0.832	-0.589	60.3%	0.529	0.832	-0.302	28.5%
index	0.918	0.918	0.000	0.0%	0.918	0.918	0.000	0.0%
coordinate	0.298	0.663	-0.365	48.7%	0.507	0.663	-0.156	16.8%

Table 7: EU rule on/off comparison across clarification costs. "off" denotes the always-commit baseline (CPA equals raw accuracy). At c=0.5 the commit threshold drops from 0.75 to 0.5, fewer scenes trigger an ask, and harm magnitudes attenuate; no listener becomes net-positive.

A.2 POSTERIOR TEMPERATURE SCALING ON THE EU RULE

Setup. If the EU rule's failure is calibration-driven, sharpening the posterior should recover its intended benefit. We apply temperature scaling $P_\tau(i) \propto P(i)^{1/\tau}$ to cached posteriors and recompute the ask-or-commit decision and CPA. Lower τ sharpens; τ=1.0 recovers the original rule. Pooled over the six datasets, n=600 per listener, speaker = *feature-canonical*, c=0.25.

Listener	τ=1.0 (rule's CPA)	τ=0.5	τ=0.25	EU off (always-commit)
direct	0.739	0.745	0.766	0.808
cot	0.242	0.430	0.689	0.832

Table 8: EU-rule CPA at three temperatures vs. the always-commit baseline. Sharpening recovers most of the cot deficit (+0.45 from τ=1.0 to τ=0.25) but no τ tested flips the sign relative to always-commit.

Result. The cot Δ_{EU} closes from -0.589 to -0.143 at τ=0.25 — a 76% recovery. Direct recovers 38%. As $\tau \to 0$ the posterior approaches one-hot and CPA approaches accuracy (the EU-off baseline), so a yet-sharper temperature converges to the baseline rather than overshoots it. *Conclusion:* calibration explains most of the cot failure but not all of it; a residual misranking effect persists.

A.3 EU RULE REPLICATION ON LISTENERS

Setup. Replicates the §5 EU-rule ablation on the four listeners — *direct*, *cot*, *elimination*, *index* — to test whether the failure is a artifact (e.g., scene-text affordance inflating posteriors). Pooled over six datasets, n=1,800 per listener at c=0.25, speaker = *feature-canonical*.

Listener	EU on	EU off (=acc)	ask%	Δ
direct	0.691	0.820	17.3%	-0.129
cot	0.254	0.766	59.4%	-0.512
elimination	0.464	0.714	39.3%	-0.250
index	0.881	0.881	0.0%	0.000

Table 9: EU-rule on/off on image-only listeners at c=0.25. The TA pattern.

Result. The hypothesis that listeners would attenuate the failure (no scene-text to inflate the cot posterior) is falsified. The EU rule's collapse is a structural property of LLM-derived posterior entropy, not a TA-listener artifact.

A.4 Speaker prompt and vocabulary engineering

Vocabulary constraint. The *feature-canonical* prompt restricts the speaker to a closed vocabulary (color, shape, size, location); *natural* elicits free-form descriptions. Holding the listener fixed at *direct-rank*(gemini-flash) at c=0.25, the constraint improves mean CPA by $+0.036$ across the six 100-scene slices, with the largest gain on `scenes_10_force` ($+0.085$). The closed vocabulary suppresses unconstrained-VLM failure modes (vague generic words, attribute hallucinations), and the effect concentrates where lexical precision matters most.

Prompt-component decomposition. We extend the vocabulary comparison by holding listener and slice fixed (pooled n=600, *direct*(gemini-flash) at c=0.25) and comparing all eight cached speaker prompts. The hypothesis was that framing-only overlays — those that change instructions without changing the input information — would land within ± 0.01 CPA of the plain *strategic-natural* prompt. The hypothesis is falsified.

Speaker prompt	CPA	ask%	mean words	Δ vs canonical
feature-canonical (rule-based oracle)	0.739	12.5%	3.4	—
natural	0.823	7.7%	5.8	+0.084
listener_aware	**0.893**	2.7%	7.1	+0.155
pragmatic	0.718	14.8%	3.7	−0.021
scene_first	0.734	13.2%	3.5	−0.005
contrastive	0.794	9.7%	6.9	+0.055
landmark	0.659	11.0%	7.8	−0.080
superlative	0.645	17.2%	4.2	−0.093

Table 10: Speaker prompts vs. *feature-canonical* (rule-based oracle) at c=0.25, pooled n=600 per row, listener = *direct*(gemini-flash). The audience-modeling overlay (*listener_aware*) is the strongest prompt; the "be pragmatic" overlay actively hurts.

The framing-only overlays span a 0.18-CPA range, comparable to the input-augmentation overlays. Audience-modeling instructions (*listener_aware*: "your listener will see the same image, write to maximize their confidence") yield the best CPA observed, while explicit pragmatic-reasoning instructions (*strategic-pragmatic*) actively hurt — possibly by inducing terser outputs.